

Semi-Rational Engineering of MddH for Enhanced H₂S Methylation Toward Whole-Cell Cascaded Biosynthesis of DMS

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Abstract

Hydrogen sulfide (H₂S) is a toxic inorganic-sulfur pollutant abundantly released from industrial manufacturing processes. Bioconversion of waste-gas-derived H₂S into dimethyl sulfide (DMS) offers a promising route for sulfur resource recycling, since DMS can act as a versatile platform intermediate linking inorganic sulfur to diverse organosulfur chemicals. Nevertheless, wild-type SAM-dependent methyltransferase MddH exhibits low substrate affinity toward H₂S and suffers from strong substrate inhibition under high sulfide concentration, restricting its application for inorganic-to-organic sulfur biotransformation. In this study, homologous sequence alignment combined with molecular docking was performed to build the

MddH-SAM-H₂S ternary complex model. Key residues within the substrate-binding pocket were identified, followed by molecular dynamics simulation and Yasara-based virtual mutagenesis to assess protein folding stability. Three candidate residues E63, E91 and T108 were selected, and corresponding site-directed mutants E63Q, E91Q and T108I as well as their combinatorial variants were constructed. Whole-cell biocatalysis combined with headspace gas (HS) was applied to characterize catalytic performance and quantify production of methanethiol (MeSH) and DMS. Several engineered MddH variants displayed improved capacity to convert inorganic H₂S into DMS. This work establishes an optimized biocatalytic module for producing the organosulfur platform molecule DMS from waste inorganic sulfur, providing a core biological element for subsequent biosynthesis of high-value sulfur-containing compounds. In the present study, catalytic characterizations were performed using crude cell lysates; reliable quantitative whole-cell assays will be pursued in further work toward the ultimate goal of whole-cell cascade biosynthesis.

Key words: hydrogen sulfide; MddH methyltransferase; semi-rational protein engineering; dimethyl sulfide (DMS); whole-cell biocatalysis

Introduction

Protein semi-rational engineering provides an effective route to break intrinsic enzyme bottlenecks without labor-intensive random-mutagenesis screening. Homologous sequence comparison reveals conserved functional regions associated with catalytic properties. Combined with molecular docking, ternary enzyme-cofactor-substrate complex models enable direct identification of critical amino-acid residues lining the substrate-binding pocket. Subsequent molecular dynamics simulation validates the reliability of ligand-protein binding poses, and Yasara-aided virtual mutagenesis calculates folding free-energy change ($\Delta\Delta G$) to filter out structurally destabilizing substitutions.

Sulfur resource recycling has attracted growing attention in industrial biomanufacturing. Waste-stream H_2S represents an abundant, cheap inorganic-sulfur feedstock. Conventional treatment strategies focus on detoxification and sulfur recovery, yet rarely realize efficient conversion into value-added organosulfur products. Dimethyl sulfide (DMS) can serve as a versatile platform intermediate bridging inorganic sulfur and organosulfur metabolism. Downstream, DMS can be further oxidized into dimethyl sulfoxide (DMSO)⁷, or be utilized as a methyl-/sulfur-donor precursor for biosynthesis of methionine, thiol-containing fine chemicals and other sulfur-bearing metabolites. Therefore, enhancing microbial conversion from H_2S toward DMS is of great practical significance⁸.

MddH, an SAM-dependent methyltransferase originating from marine microorganisms⁵, catalyzes two successive methylation reactions: inorganic H_2S is first methylated to methanethiol (MeSH), and further methylated to DMS⁶. However, wild-type MddH is limited by low affinity for gaseous H_2S and severe substrate inhibition at high sulfide concentration, which hampers its application for inorganic-to-organic sulfur conversion.

In the present work, based on homologous sequence alignment and molecular-docking analysis of MddH, three key residues E63, E91 and T108 located inside the SAM- H_2S binding cavity were pinpointed. Site-directed mutants E63Q, E91Q and T108I, alongside single-, double- and triple-site combinatorial variants, were constructed. We systematically characterized their H_2S -methylating capability and quantified yields of MeSH and DMS via headspace gas chromatography. This study focuses on improving MddH performance to convert inorganic H_2S into the organosulfur platform molecule DMS. Further downstream transformation of DMS (including oxidation to DMSO or synthesis of other organosulfur chemicals) is reserved for follow-up research. The optimized MddH variants obtained in this work supply key biocatalytic building-blocks for future whole-cell systems targeting waste-sulfur valorization. However, it should be noted that the current work focuses

on in vitro characterization using crude cell lysates, and the transition to a fully functional whole-cell biocatalytic system remains the next critical milestone.

Results

1. In Silico Stability Profiling of Site-Directed Mutants

The apo structure of the UE2 homotetramer was predicted using AlphaFold3¹¹. Molecular dynamics (MD) simulations were performed to identify residues of interest¹. The FoldX plugin³ embedded in YASARA Structure was applied to predict mutation-induced stability changes of MddH-UE2 variants^{2,4}.

To evaluate the impact of the identified substitutions on protein intrinsic stability, the FoldX plugin implemented in YASARA-Structure was applied to calculate the folding free-energy difference ($\Delta\Delta G = \Delta G_{\text{mutant}} - \Delta G_{\text{WT}}$) between each variant and the wild-type MddH-UE2, using the AlphaFold-predicted apo homotetramer as the starting structure. Positive $\Delta\Delta G$ values indicate destabilizing substitutions. As summarized in Table S1, the single substitutions E63Q and E91Q moderately destabilized the enzyme, whereas T108I was nearly neutral. Notably, the double mutant E63Q/E91Q exhibited an approximately additive destabilizing effect, while the triple mutant E63Q/E91Q/T108I showed slightly lower $\Delta\Delta G$ than the double mutant, suggesting weak epistatic compensation by T108I.

Mutation	$\Delta\Delta G$ stability(kcal/mol)	Interpretation
UE2	-	WT
E63Q	0.79	Destabilizing
E91Q	0.82	Destabilizing
T108I	0.098	Nearly neutral
E63Q/E91Q	1.66	Destabilizing
E63Q/E91Q/T108I	1.53	Destabilizing

Table S1. FoldX-predicted protein stability changes of MddH-UE2 variants.

$\Delta\Delta G$ stability(kcal/mol) = $\Delta G(\text{mutant}) - \Delta G(\text{wild-type})$. Positive values indicate destabilizing substitutions. Calculations were performed on the AlphaFold-predicted apo MddH-UE2 homotetramer structure. Variant notation E63Q/E91Q denotes the double mutant carrying both substitutions simultaneously.

Based on the above stability-prediction results from FoldX, we designed corresponding single-point and combinatorial mutants and proceeded with wet-laboratory experiments for further experimental characterization.

2. Structural Basis and Modification Strategies

2.1 Structure of MddH

Previous studies found that some bacteria can synthesize DMS without encoding the MddA enzyme in their genome. Through gene screening of such strains, researchers previously identified a novel DMS-synthesizing enzyme MddH and obtained its reference sequence. The nucleic acid sequence of MddH from *Marinobacter litoralis* strain Sw-45 was selected and translated into the amino-acid sequence. AlphaFold was used to predict the three-dimensional spatial structure of this protein, yielding the corresponding protein structural model.

Prior research indicated that this protein shares high similarity with human TMT1A and TMT1B, especially in the central and C-terminal regions. It contains the conserved central GxGxG motif and a glutamic-acid residue within the SAM-binding region (the corresponding residue in human TMT proteins is aspartic acid).



Figure 1 Results of AlphaFold spatial-structure prediction

2.2 Modification Strategy

Since quantifying the yield of reaction substrates such as DMS requires gas-chromatography detection, the associated cost and time consumption are incompatible with high-throughput screening requirements. Accordingly, conventional random mutagenesis for mutant-library construction, or direct random mutagenesis targeting the catalytic active centre, are hampered by excessive trial-and-error costs and heavy workload for wet-laboratory validation, making these approaches impractical. Furthermore, only AlphaFold-predicted structural information is available for this protein, and initial in-silico predictions may be inaccurate. To reduce the screening burden for wet-lab experiments, we adopted a combined in-silico and wet-lab workflow: in-silico predictions guide wet-lab experiments, and experimental outcomes in turn refine in-silico analyses.

Our directed-evolution strategy adopts a predict-validate-improve cyclic workflow consisting of two independent yet complementary cyclic systems. Both systems follow identical procedures in the initial design-validation phase: feasible

mutation sites are predicted based on AI-derived structural and sequence information, followed by wet-lab experimental validation. Wet-lab results are fed back into two distinct downstream cycles, for semi-rational mutant-library construction and improved in-silico prediction of mutation sites respectively.

2.2.1 Design and Validation of Beneficial Mutation Sites

The first-round prediction of beneficial sites was built on a single working hypothesis: SAM binding within UE2 is overly stable (early docking estimated -7.9 kcal/mol). Hence, weakening cofactor binding should accelerate catalytic turnover and raise DMS yield. Affinity ranking was generated via Boltz-2 simulation, based on which three mutation sites E63Q, E91Q and T108I were selected to construct single- and multi-mutants. Among these, E63Q and E91Q reside at acidic SAM-binding sites, whereas the T108I mutation is located on a side-chain that perturbs the local conformational network and thereby modulates binding affinity.

Single mutants E63Q, E91Q, T108I, double mutant E63Q/E91Q, and triple mutant E63Q/E91Q/T108I were constructed for experimental validation. Wet-lab assays revealed no obvious activity enhancement for the E63Q variant, while E91Q and T108I produced only marginal activity improvement. These experimental results were imported into the improvement module to initiate the next cycle^{14,15}.

2.2.2 Improvements Based on Validation Results

2.2.2.1 Improved In-silico Prediction Guided by Experimental Outcomes

The first-round prediction relied on a single evaluation metric (binding affinity), which failed to fully translate into expected activity gains. Furthermore, direct kinetic evidence for SAM binding (e.g., K_m determination) was not obtained in this study. To address these deficiencies, the second-round prediction underwent systematic workflow reconstruction.

Instead of relying on a single evaluation metric, mandatory sanity checks for mutated sequences were integrated in computational processing to avoid residue-numbering errors. A hard stability cutoff was implemented to replace lenient stability-filtering criteria. Three zero-shot prediction models with distinct mechanistic rationales, namely ProtSSN, ProSST and TranceptEVE, were combined with the PLACER ligand-conformational ensemble analysis module. This framework accounts for both protein stability and the actual spatial pose of SAM inside the binding pocket, rather than focusing exclusively on binding affinity. Wet-lab experiments are required to deliver sequence-validated constructs and enzyme initial-rate kinetic data, so that experimental measurements can be fed back into the model for iterative optimisation⁹.

2.2.2.2 Semi-Rational Mutant-Library Construction

In Cycle 1 of System 2, we performed pocket saturation scanning covering 45 residues within an 8 Å radius of mutations that conferred slight activity enhancement. A total of 855 single-point-mutation candidates were generated in silico for mutant-library assembly. Compared with conventional broad-scale random mutagenesis or random mutagenesis directly on predicted sites, this strategy produces a smaller mutant library, substantially lowers the workload for subsequent wet-lab validation, and improves the efficiency of the directed-evolution programme. Output data from this cycle are also fed into System 1 to support iterative optimisation.

This in-silico generated 855-member saturation-scan library has not been physically assembled or experimentally tested in the current study; it remains a future experimental option. Our near-term priority focuses on validating the Route-B short-list candidates (I46L, S65T, etc.) rather than conducting full-library screening. The

output data from this cycle are also fed into System 1 to support iterative optimisation.

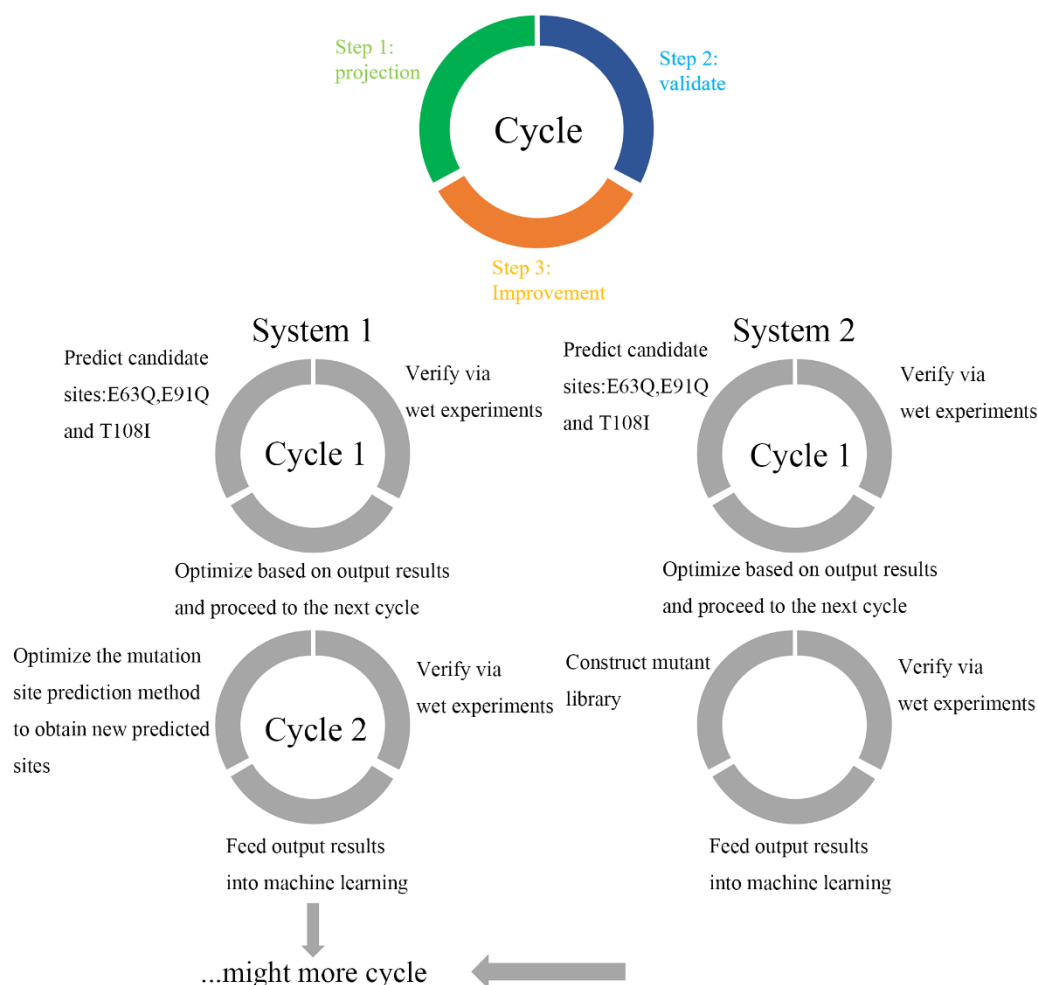


Figure 2 Two cyclic systems for the modification strategy

3.Catalytic activity of the site-directed variants

3.1 Construction of variants and activity of the crude cell lysates

Based on the computational predictions from Route A (dry lab), which hypothesized that SAM binding in UE2 is over-stabilized and weakening cofactor binding could improve DMS output, three single mutants (E63Q, E91Q and T108I) and two combinatorial variants (E63Q/E91Q and E63Q/E91Q/T108I) were commercially synthesized by Sangon Biotech. After sequence confirmation, all

variants were expressed in *E. coli*, and the crude cell lysates were used to catalyse H₂S methylation under identical conditions. The amount of DMS produced was quantified by headspace gas chromatography (HS-GC).

According to the calibration curve ($\sqrt{\text{peak area}} = 3.1854C - 25.261$), the wild-type UE-2 produced 336.4 ± 2.6 nmol of DMS. E91Q and T108I produced 353.5 ± 7.3 nmol and 344.7 ± 7.9 nmol, corresponding to 105.1% and 102.5% of the wild-type level, respectively. In sharp contrast, the introduction of E63Q caused a drastic loss of activity: E63Q yielded 16.4 ± 0.4 nmol (4.9%), E63Q/E91Q yielded 10.1 ± 2.7 nmol (3.0%), and the triple mutant yielded 29.2 ± 3.8 nmol (8.7%) (Figure 3, Table 1).

One-way ANOVA revealed an extremely significant difference among the six groups ($F(5,12) = 3979.5$, $P < 0.0001$). Dunnett's multiple comparison test against the wild type showed that E91Q produced significantly more DMS than the wild type (difference +17.1 nmol, adjusted $P = 0.0045$), whereas T108I did not differ significantly (difference +8.3 nmol, $P = 0.199$). All three E63Q-containing variants were extremely significantly lower than the wild type (differences of -320.0, -326.2 and -307.2 nmol; adjusted $P < 0.0001$) (Figure 3). A separate analysis restricted to the three variants with measurable activity gave the same conclusion: the groups differed significantly ($F(2,6) = 5.37$, $P = 0.0461$), E91Q was still significantly higher than the wild type ($P = 0.0297$), and T108I was not ($P = 0.263$).

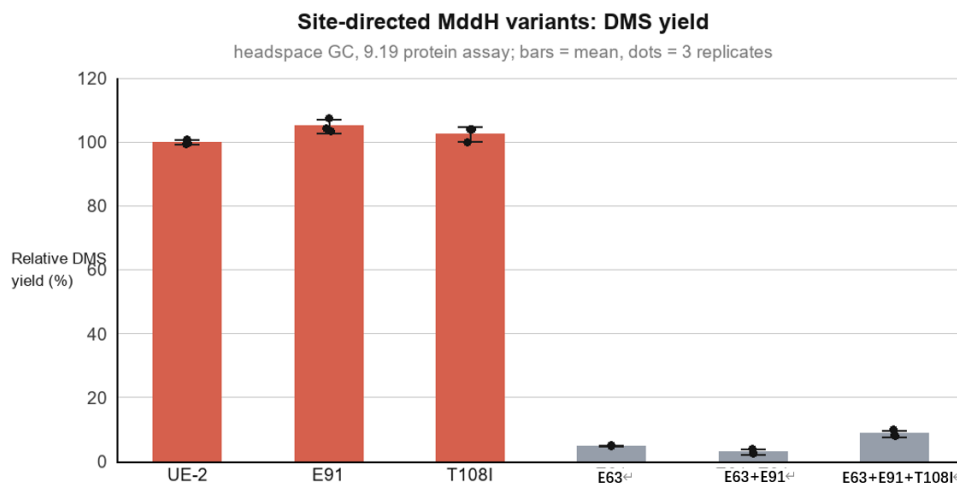


Figure 3 DMS production of MddH site-directed variants. Values are normalised to the wild-type UE-2 mean (100%); bars represent means, dots represent individual replicates (n = 3) and error bars represent SD. All variants were assayed under identical conditions and the product was quantified by headspace GC.

Strain / variant	DMS (nmol)	Relative activity (%)	CV (%)	vs wild type (Dunnett)
UE-2 (wild type)	336.4 ± 2.6	100	0.8	—
E91Q	353.5 ± 7.3	105.1	2.1	P = 0.0045 (**)
T108I	344.7 ± 7.9	102.5	2.3	P = 0.199 (ns)
E63Q	16.4 ± 0.4	4.9	2.3	P < 0.0001 (****)
E63Q/E91Q	10.1 ± 2.7	3.0	26.9	P < 0.0001 (****)
E63Q/E91Q/T108I	29.2 ± 3.8	8.7	13.1	P < 0.0001 (****)

Table 1. Catalytic activity of MddH site-directed variants. All variants were assayed under identical conditions, and the products were quantified by headspace gas chromatography (HS-GC). Data are presented as mean ± SD from three biological replicates (n = 3). Statistical significance was assessed using one-way ANOVA followed by Dunnett's multiple comparison test against the wild-type UE-2. The coefficient of variation (CV) for each variant is shown. **P < 0.01 and ****P < 0.0001 indicate significant differences compared to the wild type; ns indicates no significant difference.

Taken together, E91Q was the only substitution in this round of engineering that improved activity (+5.1%), T108I essentially maintained the wild-type level, and the E63 position was indispensable for catalysis, with its mutation abolishing more than 90% of the activity.

3.2 Replicate consistency and reliability of the data

The coefficient of variation (CV) of the three replicates within a batch reflects data reliability. The CVs of the wild type, E91Q, T108I and E63Q were 0.8%, 2.1%, 2.3% and 2.3%, respectively, indicating that the crude-lysate assay was stable and that the differences described above are not attributable to replicate fluctuation.

The two E63Q-containing combinatorial variants showed higher CVs (26.9% for E63Q/E91Q and 13.1% for the triple mutant). This is related to their product levels being close to the detection limit: when DMS production drops to 10–30 nmol, a small absolute fluctuation translates into a large relative deviation. Even so, the highest of the three replicates in these two groups (13.2 nmol and 33.5 nmol, respectively) corresponds to only 3.9% and 10.0% of the wild-type mean, far outside the replicate range of the wild type (334.5–339.4 nmol). The conclusion of a drastic loss of activity is therefore robust.

It is worth noting that the whole-cell batch was considerably less reproducible: the three wild-type replicates gave DMS peak areas of 3.36×10^6 , 3.28×10^5 and 6.45×10^6 , with a CV of 90.6% and an almost 20-fold range, and one of these values is identical to the third replicate of T108I, suggesting a recording or transcription error; by contrast, the CV of T108I itself in the same batch was only 9.9%. This indicates that the current whole-cell system still lacks adequate control over cell density, induction level and headspace sampling consistency. The quantitative conclusions of this study therefore rest on the crude-lysate data, while the whole-cell data are used qualitatively.

3.3 Differential product accumulation in E63Q variants

MddH methylates H₂S sequentially to methanethiol (MeSH) and then to dimethyl sulfide (DMS); the product profile can therefore provide preliminary insights into which step is affected in an inactive variant. MeSH and DMS were quantified in the same batch. The wild type, E91Q and T108I gave DMS peak areas of 2.28×10^7 , 2.02×10^7 and 1.95×10^7 , whereas E63Q, E63Q/E91Q and the triple mutant gave no detectable DMS peak. In contrast, these E63Q-containing variants still produced MeSH at levels comparable to the wild type (1.38×10^7 , 1.67×10^7 and 1.65×10^7 , versus 1.63×10^7 for the wild type) (Figure 3).

These observations tentatively suggest that the E63Q-containing variants largely retain the first methylation activity but are severely impaired in converting MeSH to DMS. However, it is important to note that this does not necessarily imply that E63 directly catalyzes the second step; alternatively, the mutation might induce local conformational changes in the binding pocket, indirectly affecting MeSH catalysis. Furthermore, the no-enzyme control in this batch also gave a relatively high MeSH peak ($1.24\text{--}1.77 \times 10^7$), indicating a considerable background interference in the current assay system. Therefore, the absolute assignment and precise quantification of MeSH require confirmation with stricter controls (e.g., heat-inactivated enzyme or omission of SAM), and these preliminary mechanistic interpretations should be validated by further kinetic and structural studies. Nevertheless, under identical conditions, only the E63Q-containing variants gave no detectable DMS.

Despite these limitations, these preliminary observations still served as crucial wet-lab feedback, directly prompting the dry lab to upgrade to the Route B consensus strategy, which incorporates a stability red line to avoid such structural disruption.

4. Summary

The results presented here show that (i) among the three positions selected by

homologous alignment and molecular docking, E91Q significantly increased DMS production by about 5% while T108I essentially maintained the wild-type level; (ii) E63 is indispensable for catalysis — its mutation reduced activity to below 10% of the wild type and the affected step is the second methylation, from MeSH to DMS; and (iii) simple stacking of the substitutions brought no gain, as both combinatorial variants were less active than the wild type. Further engineering should therefore focus on E91 and its surrounding interaction network, while E63 is better treated as an entry point for mechanistic study. These wet-lab findings directly inform the next cycle of dry-lab design, establishing an iterative DBTL loop.

Discussion

Semi-rational design provides an efficient route to overcome enzymatic catalytic bottlenecks. Based on homologous alignment and molecular docking, we successfully constructed the MddH-SAM-H₂S ternary complex model and pinpointed key residues including E63, E91, and T108. Compared with conventional random mutagenesis, our rational design strategy effectively narrowed the mutational search space, improved screening efficiency, and exhibited stronger predictability.

A key contribution of this study is the application of computational design to predict high-potential mutational combinations, thereby reducing experimental workload. However, the first-generation single-metric affinity prediction (Route A) did not fully translate into expected activity in wet-lab experiments. Through auditing and upgrading by the dry-lab team, we developed the Route B multidimensional consensus strategy, integrating three zero-shot predictors, a stability red line, and a ligand-aware conformational gate. Wet-lab results validated the effectiveness of this iterative strategy: the single mutation E91Q significantly increased DMS production by 5.1%, while E63Q, though accurately predicted as unstable by Route B's stability red line ($\Delta\Delta G > +1$), was further revealed by wet-lab experiments to block the second methylation step (MeSH $\rightarrow$ DMS). This closed loop of "dry-lab prediction, wet-lab

validation, mechanistic feedback, and model upgrade" demonstrates that the combination of machine learning and rational design can achieve stepwise optimization and accelerate the discovery of superior enzyme variants¹⁰.

Notably, comparable $\Delta\Delta G$ -destabilizing predictions for E63Q and E91Q yielded drastically distinct experimental phenotypes. FoldX $\Delta\Delta G$ reports global protein folding stability, yet cannot capture local active-site functional perturbation. E63Q likely disrupts key local interactions within the methylation pocket, causing near-complete loss-of-function despite moderate global destabilization; by contrast, E91Q-triggered subtle structural rearrangement confers catalytic improvement.

We developed a framework integrating rational design with machine learning for iterative optimization. This strategy was preliminarily validated through the construction and characterization of single and combinatorial mutants, which not only probed the functions of key residues but also generated a foundational dataset for model training. The dry-lab Route B consensus strategy successfully overturned the ineffective combinations from Route A, demonstrating the feasibility and self-correcting capability of this framework.

Future efforts will focus on enhancing the precision and efficiency of this cycle. We plan to systematically expand the mutant library, prioritizing the construction of next-generation candidates recommended by Route B (such as I46L, S65T, etc.). Meanwhile, we will employ standardized initial-velocity kinetic assays to replace endpoint readouts to improve data quality. To overcome current limitations, we will further optimize the prediction algorithms to better capture higher-order epistatic effects. By continuously streamlining experimental workflows and strengthening the computation-experiment iterative feedback, we will establish a robust platform for developing enzyme variants with higher substrate specificity and catalytic efficiency¹³.

Materials and Methods

Materials

Escherichia coli BL21(DE3) (provided by the instructor's lab) was used as the expression host, and the pET28 vector containing a kanamycin resistance gene was used as the selection marker.

For LB liquid medium, 10 g tryptone, 10 g NaCl, and 5 g yeast extract were dissolved in 1000 mL ddH₂O, and the medium was sterilized by high-pressure steam at 121 °C for 30 min. For LB solid medium, 10 g tryptone, 10 g NaCl, 5 g yeast extract, and 15 g agar were dissolved in 1000 mL ddH₂O, and the medium was sterilized by high-pressure steam at 121 °C for 30 min. Tryptone was purchased from Hangzhou Microbial Reagent Co., Ltd. (Hangwei), yeast extract was purchased from Oxoid (Oxoid™ Yeast Extract Powder), and agar was purchased from Qingdao Hope Bio-Technology Co., Ltd. (Hopebio). Kanamycin was supplemented at a final concentration of 50 µg/mL for selection when required.

Template DNA Extraction

A single colony was inoculated into 5 mL of LB liquid medium containing 50 µg/mL kanamycin and cultivated at 37 °C with shaking at 190 rpm until the OD₆₀₀ reached approximately 0.6. A 200 µL aliquot of the culture was transferred into a 1.5 mL EP tube and centrifuged at 12,000 rpm for 10 min to discard the supernatant. The cell pellet was resuspended in 100 µL of 1×TE buffer and thoroughly mixed until a uniform turbid suspension was obtained.

The resuspension was incubated in a boiling water bath for 10 min, followed by freezing at -20 °C for 30 min and immediate incubation on ice for 30 min. After the freeze-thaw treatment, the sample was centrifuged at 4 °C, 12,000 ×g for 10 min. The supernatant was collected as the template DNA. If not used immediately, the template

DNA was stored at -20 °C; prior to use, the sample was centrifuged again and the supernatant was collected for subsequent PCR amplification.

PCR Amplification

PCR amplification was performed using the extracted mutant plasmids as templates. Each 50 µL reaction mixture contained 10 µL of 5× GXL buffer, 4 µL of dNTP Mix, 0.5 µL of forward primer, 0.5 µL of reverse primer, 1 µL of GXL DNA polymerase, 1 µL of plasmid template, and 33.5 µL of ddH₂O.

The PCR program consisted of initial denaturation at 94 °C for 5 min, followed by 30 cycles of denaturation at 98 °C for 10 s, annealing at the primer-specific T_m for 15 s, and extension at 68 °C for 1 min per kb. After 30 cycles, a final extension was carried out at 68 °C for 10 min.

The PCR reaction mixture is shown below.

Reagent	Volume
5×GXL buffer	10 µL
dNTP Mix	4 µL
Forward primer	0.5 µL
Reverse primer	0.5 µL
GXL DNA polymerase	1 µL
Template DNA	1 µL
ddH ₂ O	33.5 µL
Total volume	50 µL

Table 2. PCR reaction mixture

Agarose Gel Electrophoresis

PCR products were separated by electrophoresis on 1% agarose gels. For the small, medium and large gels, 0.17 g, 0.34 g and 0.68 g agarose were dissolved in TAE buffer to final volumes of 17 mL, 34 mL and 68 mL, respectively. Gel stain was added at volumes of 1.7 μ L, 3.4 μ L and 6.8 μ L to the small, medium and large gels, respectively. The gel solutions were thoroughly mixed, poured into casting trays and allowed to solidify before electrophoresis.

Before loading, PCR samples were mixed with 6 $\times$ loading buffer. For 5 μ L of PCR product, 1 μ L of 6 $\times$ loading buffer was added. For a 50 μ L sample, 10 μ L of 6 $\times$ loading buffer was added. For standard wells, 3–4 μ L of sample was loaded per well, with a maximum loading volume of 5 μ L. Approximately 50 μ L of sample was loaded into large wells. DNA marker was loaded at 3 μ L per standard well or 6–7 μ L per large well. Electrophoresis was performed at 150–155 V for 20–25 min. After electrophoresis, the PCR products were visualized to confirm the presence of the target bands, and then subjected to sequencing verification to confirm the correct introduction of the mutation sites.

Induction and Sonication Lysis

The recombinant *E. coli* containing the expression vector was inoculated into LB liquid medium containing kanamycin and incubated at 37 °C with shaking until the OD₆₀₀ reached an appropriate value to obtain the seed culture. A 100 μ L aliquot of the seed culture was transferred into fresh LB liquid medium and incubated at 37 °C for 2.5 h, followed by incubation at 4 °C for 0.5 h. Protein expression was induced by adding 1 μ L of IPTG (final concentration 0.1 mM) at 16 °C for 18 h. Induced cells were harvested by centrifugation at 4 °C, 8000 rpm for 10 min, washed with fresh LB medium, and resuspended in 1 mL of PBS buffer. The resuspension was lysed by sonication on ice. The sonication program was set as 5 s of sonication with a 10 s

interval, lasting for 1 min in total, with a maximum of 3 cycles. The product was centrifuged and the supernatant was collected as the crude cell lysates.

HS-GC Analysis

The enzymatic reaction was carried out in a 200 μ L system containing crude cell lysates, SAM (1 mM), and methanethiol (MeSH, 1 mM)¹². The reaction mixtures were incubated at 28 °C for 3 h, followed by centrifugation to collect the supernatant. The supernatant was transferred into a headspace vial and sealed for HS-GC analysis.

GC analysis was performed on an Agilent 8860B gas chromatograph equipped with a headspace autosampler. Nitrogen was used as the carrier gas. The specific detector type, headspace oven temperature, injection port temperature, detector temperature, and column temperature program were set according to the standard operating procedure.

The peak areas of DMS and MeSH were integrated. The absolute product yields were quantified using the standard curve ($\sqrt{\text{peak area}} = 3.1854C - 25.261$) and expressed in nmol.

Data Collection and Preprocessing

The mutation data in this study were obtained from HS-GC experiments and stored in the "Wet-lab HS-GC data.xlsx" file, which contains the mutant names, the peak areas of DMS and MeSH, and the product yields (nmol). The peak areas were converted into absolute yields using the standard curve ($\sqrt{\text{peak area}} = 3.1854C - 25.261$). The yield of each variant was compared with the wild-type UE-2 to calculate the relative activity (%).

Computational Design

The three-dimensional structure of wild-type UE2 (MddH) was predicted using AlphaFold3. The first-round design (Route A) employed Boltz-2 to rank point mutants by binding affinity, identifying E63Q, E91Q, and T108I. The second-round design (Route B) was upgraded to a multidimensional consensus strategy. All residues within an 8 Å radius of the SAM-binding pocket were enumerated, yielding 855 single-point candidates. These candidates were evaluated by integrating three zero-shot fitness predictors (ProtSSN, ProSST, and TranceptEVE) with a ThermoMPNN stability red line ($\Delta\Delta G > +1$ kcal/mol rejected) and a PLACER ligand conformational gate. Based on the weighted consensus scores, the top-ranked variants were prioritized for future wet-lab validation, including I46L, E91Q, S65T, S44T, V40I, as well as the E91A/D/N variants designed for mechanistic probing.

The final consensus scores and candidate list were exported as a spreadsheet to guide subsequent experimental design. Based on the Route B consensus rankings, a prioritized shortlist of variants (e.g., I46L, E91Q, S65T, S44T, V40I, and E91A/D/N) was generated for future wet-lab validation. The complete ranking tables and raw computational outputs are provided in the Supplementary Material."

Data analysis and availability

HS-GC data processing

HS-GC data processing, statistical analysis, and plotting were performed using GraphPad Prism. The peak areas of DMS and MeSH obtained from HS-GC were converted into absolute product yields (nmol) using the standard curve ($\sqrt{\text{peak area}} = 3.1854C - 25.261$). Each variant contained at least three biological replicates, and the mean, standard deviation (SD), and coefficient of variation (CV) were calculated. Statistical significance was assessed using one-way ANOVA followed by Dunnett's

multiple comparison test against the wild-type UE-2. Data organization and plotting were performed using GraphPad Prism. The original data are provided as Supplementary Material.

Dry experiment project

The project code is available on GitHub

Future Perspectives

Based on the multidimensional consensus strategy of dry-lab Route B, this project has successfully predicted a new generation of high-potential candidates (e.g., I46L, S65T, S44T, V40I). Due to the time constraints of this experimental cycle, wet-lab validation of these new sites has not yet been completed, which will be the core task of the next phase.

1. Construction and Expression of Next-Generation Candidates

We will employ the established whole-plasmid PCR site-directed mutagenesis system (GXL high-fidelity polymerase, DpnI digestion) to construct the next-generation single and combinatorial mutants predicted by Route B. Subsequently, the PCR products will be directly transformed into *E. coli* BL21(DE3) competent cells. Single colonies will be picked and sent to Sangon Biotech for sequencing verification to ensure no additional random mutations. The sequence-verified positive clones will then be cultured for protein expression. When the OD₆₀₀ reaches approximately 0.6, IPTG will be added to a final concentration of 0.1 mM, followed by induction at 16 °C for 18 hours. The cells will be harvested, sonicated, and centrifuged to obtain crude cell lysates for subsequent catalytic activity assays.

2. Enzyme Kinetics Determination

To address the significant fluctuations in endpoint readouts of the current whole-cell catalysis (high CV values), future wet-lab experiments will adopt standardized initial-velocity kinetics assays, replacing the single endpoint yield detection.

3. Active Learning and the Computation-Experiment Closed Loop

3.1 Data Feedback into the Active-Learning Model

The standard kinetic data obtained above will be directly fed into the dry-lab team's active-learning module. This module will re-weight the model based on the new experimental data and evaluate the prediction uncertainty of each candidate variant.

3.2 Prediction and Recommendation of Next-Round Mutants

Based on the updated weights, the model will predict and recommend the next 16–32 optimal variants, fully completing the Design-Build-Test-Learn (DBTL) closed loop. Through this continuous iterative optimization, MddH variants with significantly improved catalytic efficiency are expected to be obtained.

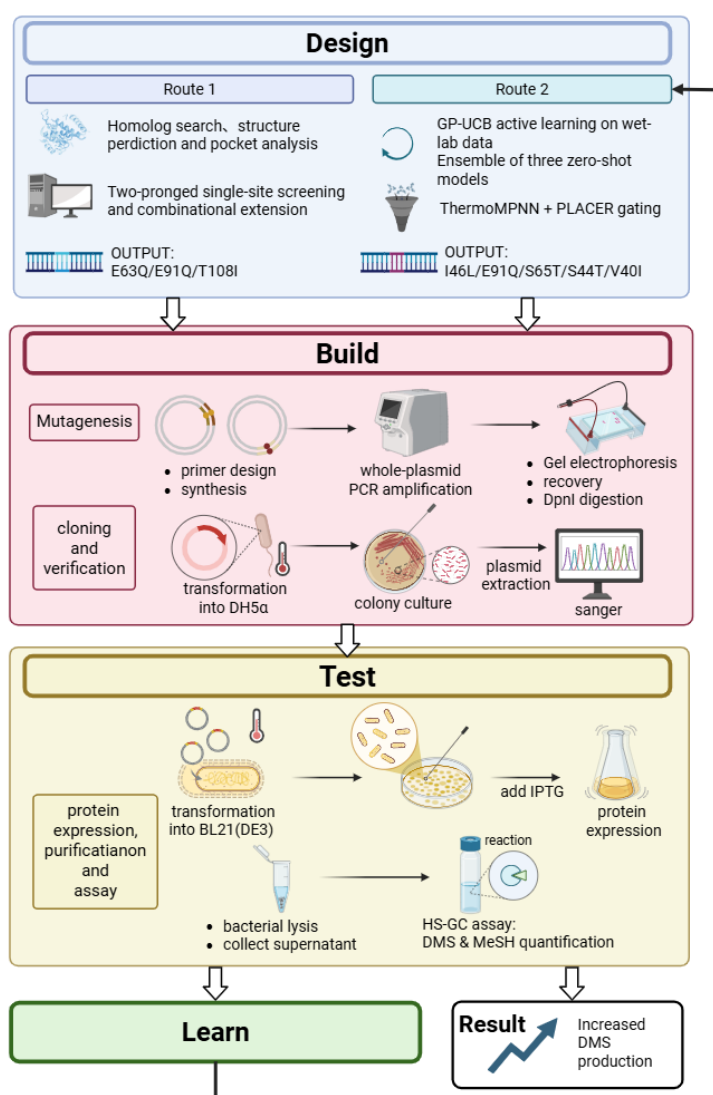
4. Cascade System Assembly and Whole-Cell Catalysis Application

4.1 Co-expression Assembly of MddH and Tmm

Ultimately, we hope to co-express the optimized MddH variants with the monooxygenase Tmm in the host to reconstitute the complete $\text{H}_2\text{S} \rightarrow \text{MeSH} \rightarrow \text{DMS} \rightarrow \text{DMSO}$ multi-enzyme cascade pathway.

4.2 Performance Evaluation of Waste Sulfur Valorization

We will quantitatively detect the yield of the final product DMSO of this cascade system and systematically evaluate the overall performance of the whole-cell catalytic system, aiming to achieve the biotransformation of waste inorganic sulfur into high-value organosulfur platform molecules, providing biological elements for the recycling of industrial waste sulfur resources.



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Figure 4. Overall DBTL workflow of the project. Route 1 (left) represents the first generation of computational design (E63Q/E91Q/T108I), which has been experimentally validated in this study. Route 2

(right) represents the upgraded multidimensional consensus strategy((I46L/E91Q/S65T/S44T/V40I). The Build and Test steps for Route 2 (indicated by the pathways leading back to the Design cycle) are currently ongoing; their corresponding wet-lab validation is planned as the next phase. The learning module (bottom) feeds experimental data back to update the model, closing the iterative DBTL loop.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT, Codex, Qwen, DeepSeek, and Doubao in order to assist with literature collection, code debugging, and language refinement. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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